SYSTEMIC FAILURES IN ISRAEL'S TAMIR-LED PACKAGING RECYCLING: STRUCTURAL AND INSTITUTIONAL CONFLICTS

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Введение. TAMIR - Declaration vs Reality

TAMIR Association (The Israeli Recycling Corporation, תאגיד מחזור יצרנים בישראל בע"מ, חל"צ) was created as a key tool for implementing the principle of Extended Producer Responsibility (EPR) in the packaging sector in Israel (1), (2), (3). In the formal logic of the model, producers and importers of packaging finance the collection, sorting, and recycling of waste from their products, compensating for costs previously imposed on the population through municipal taxes. Payment for packaging disposal by producers should partially finance city authorities, reduce waste landfilling, and bring the country closer to a circular economy (1), (2), (3).

This construction was enshrined in the Packaging Treatment Regulation Law (חוק להסדרת הטיפול באריזות, 2011). According to the law, producers and importers of goods in packaging are obliged either to independently ensure achievement of established collection and recycling targets, or to transfer these obligations to a recognized operator - "גוף מוכר". In practice, the overwhelming majority of companies chose the second path. Since 2011, TAMIR corporation became such an operator, and essentially it turned into the central link of the entire Israeli EPR system for packaging (2), (3). In 2021, the Ministry of Environmental Protection extended its status for another five years until the end of 2026, thereby effectively consolidating its monopoly position in this field (3), (4).

If one relies on the official narrative reproduced on the TAMIR website, in government documents, and in international reviews including OECD Environmental Performance Review: Israel 2023, the picture appears consistent and even convincing. Producer pays: companies contribute "treatment fees" (דמי טיפול) depending on the weight and type of packaging they place on the market, and these payments should cover all main system expenses - from containers and logistics to sorting and recycling (1), (2), (3), (5). Municipality should not be the main payer. In EPR logic, its role is limited to providing sites, participation in organizing urban infrastructure, and informing residents, while the financial burden should be compensated through TAMIR payments (2), (3), (6).

No less important part of this official image are the recycling indicators. In TAMIR reports and Ministry of Environmental Protection presentations, it is emphasized that the system not only works but demonstrates high efficiency. For example, according to the Packaging Law, the overall recycling target for Israel is set at no less than 60% of the mass of packaging waste introduced to the market (see TAMIR explanation for producers and importers: Ynet, 2025). In its public communications, TAMIR emphasizes that achieved recycling rates for declared packaging volumes exceed this standard, however the calculation structure and completeness of coverage of the entire packaging waste stream remain opaque to external observers.

It should also be noted that these materials particularly highlight paper and cardboard, metal, and wood fractions - i.e., precisely those categories where statistics look most favorable. In a broader strategic context, TAMIR is positioned as one of the flagship tools of Israel's ecological modernization: as a system that should help the country reduce waste landfilling, strengthen recycling, integrate into international initiatives like EXPRA, and approach the European model of secondary raw materials management (1), (3), (5), (7).

At the declaration level, all this indeed aligns well with international rhetoric. In OECD reviews, the packaging EPR system explicitly appears as one of the key elements of Israel's "green growth" policy alongside the landfill tax (היטל הטמנה) and Cleanliness Fund (קרן הניקיון) (5), (7). Government reports prepared within the framework of interaction with OECD and EU also emphasize that Israel allegedly moves in line with European practice precisely thanks to the packaging law and creation of a centralized industry system in the form of TAMIR (5), (7).

However, this neat picture noticeably changes as soon as we move away from declarations and look at how the system operates in reality. The gap between how TAMIR and EPR are described on paper and how they function in practice turns out to be neither incidental nor marginal. On the# Introduction. TAMIR - Declaration vs Reality

The TAMIR Association (The Israeli Recycling Corporation, תאגיד מחזור יצרנים בישראל בע"מ, חל"צ) was created as a key instrument for implementing in Israel the principle of extended producer responsibility - Extended Producer Responsibility, EPR - in the packaging sector (1), (2), (3). In the formal logic of the model, producers and importers of packaging finance the collection, sorting, and recycling of waste from their products, compensating for costs previously imposed on the population through municipal taxes. Payment for packaging disposal by producers should partially finance local authorities, reduce waste landfilling, and bring the country closer to a circular economy (1), (2), (3).

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We also note that these materials particularly highlight paper and cardboard, metal, as well as wood fractions - that is, precisely those categories where statistics look most favorable. In a broader strategic context, TAMIR is generally positioned as one of the flagship instruments of Israel's ecological modernization: as a system that should help the country reduce waste landfilling, strengthen recycling, integrate into international initiatives like EXPRA, and approach the European model of secondary raw materials management (1), (3), (5), (7).

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However, this coherent picture begins to noticeably change as soon as we move away from declarations and look at how the system operates in reality. The gap between how TAMIR and EPR are described on paper and how they function in practice turns out to be neither incidental nor random. On the contrary, it has a distinctly structural character.

First of all, already in official documents of the Ministry of Environmental Protection and State Comptroller reports (מבקר המדינה), it is acknowledged that the packaging EPR system covers only part of the actual packaging waste flow in the country. Estimates vary here, but overall we are talking about only 28–55% of the total volume of generated packaging (1), (6), (8). In other words, a significant part of packaging does not enter at all the contour along which official reporting is then built. Added to this is another chronic problem: a noticeable number of producers and importers effectively remain outside the system. These free riders (יצרנים ויבואנים שלא ממחזרים אריזות כחוק) do not participate in recycling financing to the proper extent, which means the "producer pays" model does not operate in full volume from the very beginning (6), (9), (10). State law enforcement remains fragmented: the number of inspections and volume of actually collected fines are substantially below the level possible under systematic control, and automatic data integration between customs, tax authorities, and corporate registries still does not exist (6), (9), (10).

No less significant is the question of what exactly official recycling indicators measure. Independent reviews, including InfoSpot materials, Haaretz/TheMarker publications, and research organization analyses, emphasize: high recycling percentages are calculated not from the full actual packaging waste flow, but from the so-called "declared base" - that is, from the packaging volume reported by system participants (1), (6), (8), (9), (11). At first glance this may seem a technical detail, but in fact precisely here passes one of the main boundaries between declaration and reality. Because 90–100% recycling rates for declared volumes can quite coexist with a different picture: a significant part of packaging does not pass through the TAMIR system at all. Moreover, entire fractions effectively drop out of statistics - either because they are produced by non-participating producers, or because they are not collected and sorted within TAMIR infrastructure and go with mixed waste to landfills and transfer stations. This is particularly noticeable in the case of some plastics and composite packaging (1), (6), (8), (9), (11).

But for this paper, one more aspect is fundamentally important - the financial one. In the official EPR construction, it is assumed that the main burden of expenses falls on producers, while municipalities only help organize separate collection. However, analysis of municipal budgets, tender documentation, and waste management practices in large cities such as Haifa and Tel Aviv, as well as in a number of medium-sized municipalities, shows that in reality the situation looks different. Standard TAMIR payments do not cover cities' real costs for organizing and maintaining the system, especially where there are complex urbanistic and topographical conditions - mountainous terrain, dense development, socially heterogeneous population (6), (8), (12). As a result, cities are forced to compensate the deficit from general sanitation and MSW removal budget items, and thus from arnona paid by residents and local business regardless of whether they produce packaging or not (8), (12).

In some cases this gap becomes particularly noticeable. Thus, in Jerusalem and a number of peripheral areas, TAMIR infrastructure was deployed with substantial delay relative to the moment when the law was already formally in effect. At the same time, service quality and frequency of orange container collection often lagged behind standards of regular mixed waste collection, which caused both resident complaints and critical media publications (1), (6), (11), (13). And here it is especially clear: between the normative model and its actual implementation exists not just an administrative gap, but a deep institutional misalignment.

Against this background, international reviews by Israeli authorities themselves and OECD, including Post-Accession Progress Report on Waste Management and OECD EPR Guide, are particularly indicative. They record that, despite the existence of EPR system and formally high recycling rates for declared volumes, Israel still remains one of the OECD countries with very high waste landfilling levels. The share of MSW recycling overall also remains substantially below the OECD average (5), (7), (14). From this follows a rather harsh conclusion. Either the EPR system operates on too narrow a stream segment and practically does not affect the main MSW mass. Or economic incentives - including landfill tariffs, Cleanliness Fund resource use logic, and gate fees structure - are arranged so that recycling still proves less profitable than landfilling.

Precisely therefore already in the introduction one can fix the key paradox of the entire system. In official discourse, TAMIR appears as the central mechanism for implementing the "polluter pays" principle and as an important instrument for transition to circular economy. In operational reality before us is a much narrower, partially underfunded waste management system segment, a significant part of whose costs is in fact shifted onto municipalities and taxpayers, while declared efficiency indicators are built on incomplete and methodologically limited data base.

Hence follows the task of this analytical paper. Its goal is not to once again reproduce formal reports or PR communications, but to show how precisely the financial and institutional architecture of TAMIR is arranged in the real Israeli waste management system. At the center of attention will be three interrelated questions: 1) how this construction works in practice; what structural conflicts distort its functioning - first of all in relations with the deposit system, landfill tariff policy, and Cleanliness Fund mechanisms; and why actual distribution of costs and benefits so noticeably diverges from the declared EPR principle. In other words, the main focus is to what extent the TAMIR system in its current form does not so much resolve the packaging waste crisis as reproduce it in a new institutional form.

1. Architectural flaw: conflict between TAMIR and deposit system

Israeli packaging waste management system is structured not as a single and internally consistent mechanism, but as a combination of two parallel regimes, each operating according to its own logic. On one hand, there exists the deposit system for beverage containers (פיקדון), historically associated with ELA (אגודת א.ל.ה.) and new reverse vending machine operators (15), (16), (17). On the other hand operates the Extended Producer Responsibility system for packaging, implemented by TAMIR as the sole recognized operator under the Packaging Law (2), (3).

At the formal level, this construction seems quite reasonable. It is assumed that the deposit system is responsible for beverage bottles and cans, while TAMIR handles all other packaging. That is, one system supposedly complements the other. However, in reality between them arises not just technical overlap, but a deep systemic conflict. And the problem here is not in individual shortcomings, but in the very architecture of the model: it initially distributes between the two regimes not only different waste fractions, but also different levels of economic attractiveness.

Deposit system operates based on the Deposit Law on Beverage Containers (חוק הפיקדון על מיכלי משקה). Its principle is simple: the buyer pays a fixed deposit when purchasing a beverage, and then receives this money back when returning the empty container (15). But behind this apparently simple scheme hides an important economic effect. The system covers plastic bottles and metal cans up to 1.5 liters from the moment of launch, and from December 1, 2021 also extended to large plastic bottles with volume from 1.5 to 5 liters (15), (18). The size of the standard deposit is 0.30 ₪per container unit (15), (18). According to various estimates, 2.5 to 3 billion packaging units pass through this channel annually, of which about 65–70% are PET bottles (15), (19), and the return rate in different years and regions stays approximately in the range of 70–85% (19), (20).

For consumers this looks like a convenient way to return a bottle and get money back. But for waste economy the meaning is much more serious. Deposit system collects containers in relatively clean, standardized form, with minimal organic contamination and foreign impurities, through stores and RVM machines (15), (19). And since PET and aluminum have stable market demand, precisely these materials become the most valuable and liquid elements of the packaging stream (19), (20). Not coincidentally, in international documents deposit systems are described as high value stream capture mechanisms - mechanisms extracting from the general waste stream the highest quality and most profitable materials for recycling (20).

Precisely here begins the problem for TAMIR. As EPR operator for packaging, it works with everything remaining outside the deposit system: packaging from dairy products, oils, sauces and household chemicals, composite materials like TetraPak, films, bags, trays, non-deposit metal packaging, paper, cardboard and glass not covered by deposit (2), (3), (6). Theoretically this too can be presented as logical division of responsibilities. But practically this means that what enters the TAMIR system is not simply "other part" of packaging, but an inherently less profitable, more contaminated, more difficult to sort, and less attractive for secondary raw materials market stream.

TAMIR containers receive mixed packaging with food and beverage residues, large number of incorrectly discarded items, high share of materials that are either difficult to sort or nearly impossible to recycle without substantial costs. This also concerns non-deposit glass, which is unprofitable for consumers to return, and expensive for municipalities and operators to collect and transport (1), (6), (21). Therefore TAMIR receives not a neutral packaging waste stream, but a stream from which the most valuable components have been pre-extracted.

Official and independent sources agree that through the so-called "orange stream" passes only a limited part of packaging. In 2022, through it was collected about 6.2% of the mass of packaging falling under the Packaging Law (6), (21). For plastics recycling rates remain relatively low - around 30% of declared waste mass (1), (6). For glass the situation is even weaker: in 2022–2023 recycling rate was only in the range of 4–8% (1), (6), (21). In other words, TAMIR system operates on the most difficult, least profitable, and at the same time most costly part of the packaging stream.

From this follows the next, already financial effect. When the most liquid fractions leave the overall stream, the economics of the remaining container system begins to deteriorate almost automatically. Before expanding deposit to large bottles at least part of PET still remained in TAMIR and municipalities zone. After the reform even this segment largely moved to a separate regime (18), (22). For TAMIR and cities this meant loss of the last relatively profitable plastic fractions. As a result, the specific weight of loss-making raw materials increased, and the cost of sorting and recycling drifted even further from potential revenues (1), (6), (21).

Therefore the conflict between deposit system and TAMIR cannot be considered secondary. From state perspective one can report growth of bottle returns and present this as environmental success (18), (22). But looking at the entire system, it becomes clear that this success is achieved through stream redistribution: all valuable concentrates in one regime, all

complex and expensive - in another. Meanwhile TAMIR continues paying municipalities fixed standard compensations that do not reflect real deterioration of stream composition. "דמי טיפול" tariffs are not automatically recalibrated to this new situation (6), (21), (23). As result grows hidden deficit which cities are forced to cover from their own budgets.

For municipalities this is particularly sensitive. They remain obliged to maintain collection infrastructure, maintain sites, service containers, organize removal, control sanitary condition of accumulation points. But monetary potential of most attractive fractions goes to deposit system. In other words, cities retain significant part of expenses but lose even those limited benefits that could partially compensate separate collection costs. In practice this means that share of costs covered under real EPR principle turns out lower than declared in official model, and the difference is shifted onto municipal budgets and through them - onto residents (6), (21), (23).

Here also manifests conflict of interests between system participants. For deposit scheme operators it is beneficial for deposit container volume to grow: this means expanding access to valuable raw materials, increasing turnover and, within certain limits, growing revenues from unclaimed deposits (15), (19), (20). For TAMIR, conversely, it is important to preserve sufficiently broad and economically viable base of packaging waste so the system does not turn into servicing exclusively loss-making fractions. Municipalities are primarily interested in reducing overall MSW management costs and stable operation of urban sanitation infrastructure. Thus one and the same step - for example deposit expansion - can be success for one system part and simultaneously blow to another.

Particularly clearly this architectural flaw manifests in glass example. Precisely glass shows that the problem is not reduced to plastic bottles and not exhausted by PET dispute. Deposit system covers glass beverage containers only partially and mainly where economically justified (15), (18). Remaining glass packaging - jars, bottles from sauces, preserves and other food products - falls into responsibility zone of TAMIR and municipalities, primarily through purple containers (1), (3), (21).

But this model has extremely weak economic base. Glass has relatively low EPR tariffs, consumer receives no direct incentive to return it like deposit containers, the material itself is heavy and fragile, its transport is expensive, and income from secondary glass on Israeli market is limited (1), (6), (21). Meanwhile purple container infrastructure is deployed unevenly, and expansion targets to 2026 still not achieved (1), (6), (21). Therefore extremely low glass recycling rate - only 4–8% looks not like random deviation, but quite logical result of entire system design (1), (6), (21). Glass here is a symptom: deposit system did not truly integrate it, TAMIR received heavy and low-profitability stream, municipalities - another costly infrastructure obligation.

Against this background particularly noticeable is another weakness of Israeli model: absence of unified systemic management of packaging waste. Country lacks institutional center that would simultaneously coordinate deposit system, EPR through TAMIR, and landfilling economic parameters including landfill tariffs and Cleanliness Fund mechanisms. No mechanism also exists for redistributing rent from profitable streams - primarily from deposit

PET and aluminum - in favor of chronically loss-making fractions such as mixed polymers, dairy packaging or non-deposit glass.

Ministry of Environmental Protection operates in logic of separate laws and separate regulatory regimes. One law regulates deposits, another - packaging, separately exist landfilling tax and Cleanliness Fund mechanisms. As result each actor defends primarily own niche: deposit operators - theirs, TAMIR - its own, landfill operators - theirs, municipalities - theirs (5), (6), (7), (18), (22), (23). From this does not emerge coherent packaging waste policy. Rather emerges system of competing regimes where success of one segment can be maintained at expense of weakening another.

Therefore at polluter pays principle level current model appears internally contradictory. Formally beverage and other packaging producers participate in system financing, but actually they pay less than would cost truly complete, honest and integrated waste management model, especially considering real urban expenses. Meanwhile most profitable elements - unclaimed deposits and highly liquid secondary raw materials - concentrate in relatively narrow segment of deposit operators and associated recyclers (15), (19), (20), (23). Everything else - primarily expensive logistics, weak fractions and infrastructure deficit - remains dispersed across TAMIR system and municipal budgets.

As result Israel gets not integrated packaging policy, but constant friction between regimes. Easy to demonstrate growth of deposit container returns: this is visible, understandable and politically advantageous. Much harder to show honest recycling growth across all packaging fractions and real waste flow reduction to landfills when basic system economics undermined from start. Precisely in this consists main architectural flaw: deposit system and TAMIR exist not as mutually reinforcing elements, but as structures between which distributed not only functions, but also resources of unequal value. And in this sense Israeli model noticeably lags behind European PPWR logic where packaging considered as single chain - from design and reuse to recycling and responsibility allocation (7), (20), (24).

2. Economics of "invisible subsidy": Municipal gap

One of central declarations of Israeli Extended Producer Responsibility model consists in that EPR system should relieve municipalities of main financial burden of packaging waste management. In this logic producers and importers through TAMIR not simply participate in recycling financing but actually assume costs of collection, sorting, and servicing of packaging stream. Cities meanwhile assigned rather organizational role: provide infrastructure, integrate separate collection into existing sanitation system, and receive compensation for this. Formally everything looks quite consistently: private sector pays for own waste, and burden on local budgets and accordingly on residents should decrease (2), (3), (6). However analysis of public reports, tender documentation, OECD materials, and independent reviews shows that in practice a different picture has formed. TAMIR system to a significant extent rests on hidden financial support from municipal budgets which is not directly called subsidy but factually is exactly that (1), (5), (6), (7), (21), (23).

On paper compensation mechanism arranged quite simply. According to Packaging Law and standard agreements between TAMIR and municipalities, corporation undertakes either to install and service infrastructure itself or do this through approved contractors, and also pay

cities standard compensation for collection and removal of packaging waste from TAMIR containers. Sometimes this same formula partially includes costs of educational work and integration of separate collection into city waste management schemes (2), (3), (6), (21). Compensation calculated by fixed rates - usually per ton of collected packaging, and in some models also taking into account number of containers or serviced population (6), (21), (23). Precisely here arises key discrepancy: standard looks like neat formula but city economy almost never lives by averaged formula.

Problem begins with that these compensations poorly account for real geography and urban form of specific cities. Haifa, Jerusalem and other cities with complex terrain bear far from same costs as flatter and more compact municipalities. In practice this means additional trips, lower truck capacity on complex routes, accelerated equipment wear, more expensive logistics, and in old neighborhoods - also need to use non-standard transport or more manual labor (1), (6), (12), (13), (25). Added to this are costs which in abstract model generally look secondary but in city management prove constant: container damage, vandalism, arson, theft, graffiti. As result real cost per ton of packaging waste for city - if counting not only removal but also wages, fuel, equipment depreciation, contract management, and infrastructure repair - proves noticeably higher than price which TAMIR builds into its standards (1), (6), (12), (13), (25).

Gap exacerbated also by that waste stream itself becomes ever less profitable for municipalities. As already shown in first part, deposit system takes from overall stream most valuable fractions - primarily PET and aluminum, especially after deposit extension to large bottles (15), (18), (19). For cities this means simple but unpleasant thing: what potentially could generate income or at least improve separate collection economics gradually goes to another regime. Meanwhile in TAMIR system remain most complex, dirty, and loss-making fractions - mixed plastics, dairy product packaging, non-deposit glass, and other low commercial value materials. But TAMIR standard rates do not automatically adjust to this. They fail to keep up neither with stream composition change nor with rising fuel prices, wages, landfills, and equipment maintenance (6), (21), (23), (25). Therefore each step of deposit expansion without simultaneous revision of TAMIR financial model factually increases deficit on municipality side.

There is also another layer of costs which in such schemes almost always underestimated - administrative. Municipality spends money not only on trucks, containers, and workers. It spends them on preparing and supporting tenders, contractor control, inspections, resident complaint processing, information campaigns, coordinating different waste streams, and internal analytics without which system simply ceases being manageable (5), (6), (7), (12), (13). In other words, TAMIR pays for hauled ton but almost does not pay for very management system making this ton possible. Meanwhile separate collection is always not only physical but also organizational infrastructure. The more complex system becomes, the more containers, different routes, separate reporting forms, and conflicting requirements from regulator and operators it contains, the higher these transaction costs grow. Precisely they particularly poorly visible from outside but precisely they often make entire scheme for city chronically loss-making (5), (6), (7), (12), (13).

Added to this are capital investments which formally not counted as part of "operational" cost of separate collection but in reality fall precisely on city budget. Talking about preparing sites, ground reinforcement, access arrangement, installing fences, cameras, lighting, and sometimes purchasing part of containers within improvement programs (1), (6), (12), (13). Even when TAMIR formally provides containers, city still cedes valuable street space for them, finances point relocation or redesign, bears costs of cleaning around problematic sites and eliminating fly-tipping. All this almost not reflected in standard compensation although precisely from such "details" adds up real system price for municipality. Therefore hidden subsidy arises not in some one budget line but dispersed across multitude of expenses which official model prefers not to link directly with TAMIR (1), (6), (12), (13), (25).

When talking about scale of this hidden subsidy, exact figures naturally differ from city to city. But even cautious estimates found in expert discussions, municipal documents, and specialist publications show talking not about insignificant sums but tens of millions of shekels per year for large city (1), (5), (6), (7), (12), (13), (21), (23), (25). For Haifa or Be'er Sheva scale municipalities annual gap between real costs of servicing separate packaging collection and TAMIR compensations can reach 10–20 million ₪if accounting not only direct removal but also infrastructure, overhead costs, and associated system management (5), (7), (21), (23), (25). In Tel Aviv scale metropolis absolute gap can be even higher although there it more heavily dissolved within general sanitation contracts making it harder for external observer to understand which expense part relates precisely to packaging and which to city sanitation overall (7), (21), (25), (26). Existence of such gap indirectly indicated also by cities' own complaints about compensation insufficiency, sanitation department heads' statements that separate collection idea good but its economics "does not add up," and constant need to cover missing expenses from general budget items primarily arnona receipts (5), (6), (7), (12), (13), (15), (21), (23), (25), (26).

If trace how money actually moves in this system, becomes clear that "producer pays" formula works only partially. Producers and importers really pay TAMIR "דמי טיפול" calculated based on mass and type of declared packaging (2), (3), (21), (23). TAMIR in turn pays contractor services, pays municipalities standard compensations, and finances own apparatus and public communication (1), (3), (6), (21). But then appears next level: municipalities from own budgets cover difference between real costs and received compensation, and also assume infrastructure and administrative expenses which EPR system factually does not pay (5), (6), (7), (12), (13), (25). And only after this into scheme enter residents and local business who through arnona and other local payments finance this hidden deficit and often additionally bear indirect costs in form of overflowing containers, deteriorating sanitary situation in individual areas, and overall pressure on city budget (12), (13), (25), (26).

Precisely therefore here appropriate to speak about substitution of original principle. Formally system built on polluter pays idea but in reality increasingly obtained something closer to resident and local business pay. Producer pays less than would cost really complete system not relying on hidden municipal subsidization; otherwise both "דמי טיפול" tariffs and overall volume of producer obligations would be substantially higher. Municipalities and residents pay more than EPR logic assumes because forced to cover gap between normative

model and real cost of its execution as well as consequences of architectural conflict between TAMIR, deposit system, and broader landfilling economy (2), (3), (5), (6), (7), (21), (23), (25), (26). Meanwhile in public and political discourse this gap almost not articulated. Cities unprofitable to openly conflict with TAMIR and producers, Ministry of Environmental Protection more advantageous to show reform success and high recycling percentages, and residents generally do not see in their bills separate line directly showing them that part of their arnona factually goes to co-financing EPR system.

In this consists main conclusion of second section. "Invisible subsidy" is not side defect and not private administrative error but systemic element of current configuration. While standard compensations detached from real city economy, while most valuable fractions go to deposit contour, and infrastructure and management costs remain on municipality side, Extended Producer Responsibility principle realized only partially. Outwardly system may look like EPR but internally continues relying on very same municipal and tax base which by idea should have relieved.

3. Systemic inefficiency and "blind spots"

Even if one proceeds from the official logic in which TAMIR is considered a successful EPR operator, too much in this system begins to fall apart upon closer examination. Formally, target indicators are met, reporting looks impressive, and the very construction is presented as proof that Israel is moving toward a circular economy. But if one looks not at declarations, but at the system's internal structure, it becomes clear that its architecture contains a whole series of built-in blind spots. Precisely these allow a significant part of the market to avoid responsibility, and the regulator to rely on statistics that create the impression of success but increasingly poorly reflect the real ecological picture (1), (6), (8), (11).

One of the most serious problems is the scale of evasion from system participation. The very EPR principle assumes a simple thing: any producer or importer placing packaging on the market must pay for its subsequent disposal. In other words, the system works only if market coverage is sufficiently complete and no one has the opportunity to stay aside, shifting their costs onto others. However, in Israeli practice a stable segment of "free riders" has formed - companies that either did not enter the system at all or underreport volumes of declared packaging (9), (10). According to various estimates, including InfoSpot materials and Ministry of Environmental Protection reports, TAMIR covers only 28% to 55% of the total volume of packaging waste generated in the country (1), (8). This means not just statistical undercounting, but something much more serious: a significant part of packaging exists from the beginning outside the financial and regulatory responsibility on which EPR should actually be built.

Particularly important is that we are talking not only about formal non-participation of individual players, but also about the weakness of the control mechanism itself. The Ministry of Environmental Protection acknowledges that oversight of importers and small producers remains fragmented. To date, there is no full automatic cross-check between customs data, tax reporting, and TAMIR registries, which means thousands of companies can exist for years in a gray zone without bearing full responsibility for packaging they place on the market (6),

(9), (10). As a result, not just an administrative shortcoming arises, but direct economic distortion. Those who do not pay gain competitive advantage: they do not include environmental fees in product price, while conscientious system participants are forced to bear growing burden. Essentially, part of TAMIR infrastructure costs is shifted onto those producers who remain within the system, and their "דמי טיפול" tariffs become a tool for patching holes formed due to incomplete market coverage (21), (23).

From this directly follows the second problem - how exactly official success statistics are constructed. The statement about "97% recycling" looks impressive exactly until the question arises: 97% of what exactly are these calculated? In reality, it concerns not the entire actual packaging waste flow in the country, but only the volume of packaging declared by TAMIR system participants (1), (6), (21). And this changes the entire meaning of the figure. Because if a significant part of the market did not enter the system at all, then even 100% recycling of declared volume would not mean 100% recycling at country scale. It would mean only a high indicator within a narrowed and pre-selected base. Precisely therefore it is appropriate to speak here not just of controversial methodology, but of a statistical trap. Formally TAMIR can report successes, but this success exists within a limited, administratively delineated contour that does not coincide with real ecological picture (6), (8), (11).

This methodological gap is particularly noticeable if comparing TAMIR reports with overall waste statistics in Israel. While the corporation demonstrates high recycling percentages from declared base, the country overall remains one of the most landfill-dependent in OECD space: about 80% of waste still goes to landfills (5), (7), (14). In other words, TAMIR success does not convert into comparable improvement at entire system level. This is the main sign that we are dealing not with full systemic effect, but with local statistical bubble existing alongside much less favorable reality.

Another blind spot is related to which fractions the system can actually recycle. TAMIR shows most convincing results where the stream is relatively simple and economically clear: cardboard, paper, metal. But as soon as it comes to complex, heavy, contaminated or low-liquidity materials, the system starts failing (1), (6), (21). Glass situation is particularly indicative. Recycling rate of packaging glass - jars and bottles without deposit - remains at catastrophically low level, in 4–8% range (1), (6), (21). The reason is not one. Consumer has no direct financial incentive to return such containers, heavy and fragile material logistics is expensive, and standard TAMIR payments do not fully compensate cities these costs (12), (25). As result glass exists in the system as chronically problematic fraction for which formally there is infrastructure, but no really working economic model.

Similar situation is observed with complex plastics. Excluding PET, plastic packaging recycling rates remain low - about 30% (1), (6). Multi-layer packaging, dairy plastic, contaminated containers and other technically difficult fractions often prove either economically unprofitable for sorting or generally unrecyclable at existing capacities. Then they go to landfills already after sorting - under guise of so-called "residues" unfit for recycling (6), (8), (11). This is particularly important because at public communication level often created impression that separate collection automatically means recycling. In practice

between container and actual recycling lies long chain of technical and economic filters, and far from everything that entered the system passes it to the end.

No less indicative is that in this zone of systemic failures almost no innovation incentives are visible. TAMIR monopoly status and fixed tariff structure do not create strong pressure toward deep recycling of difficult fractions. Easier for system to meet plan through relatively convenient materials than invest in solving glass, complex plastics or contaminated stream problems (5), (7), (21). In this sense current model not so much pushes market toward technological renewal as consolidates reporting-convenient selectivity: recycle what is easier to recycle anyway, and not too actively intervene where recycling requires additional investments, new capacities and riskier solutions.

To these material limitations adds also spatial inequality. Even where system formally exists, its infrastructure is distributed extremely unevenly. Coverage by orange containers, service quality and implementation speed noticeably differ between country center, Jerusalem, periphery and individual urban areas (12), (13), (25). For residents this means very simple thing: access to separate collection turns out not universal right, but variable depending on geography. People can buy goods from producers who formally pay fees to TAMIR, but for years not have nearby actually accessible and stably serviced separate collection point. Thereby system creates zones of peculiar ecological alienation: responsibility as if declared at national level, but actual access to its implementation distributed extremely unevenly.

Particularly acutely this manifests in so-called "last mile" problem. TAMIR is responsible for overall system contour, but does not bear full responsibility for what happens around specific container. Meanwhile precisely there arises significant part of real problems: overflow, garbage around site, fly-tipping, unsanitary conditions, resident complaints (12), (25), (26). All this becomes municipality concern which is forced to spend resources on cleaning, response and sanitary order restoration without receiving equivalent compensation from corporation. And here again manifests same logic we already saw in previous sections: formally system looks centralized and orderly, while real costs are blurred and pushed down to city level.

Against this background not surprising that population trust also suffers. When residents see overflowing containers, dirty sites or read media reports that separately collected waste then still "dumped in one pile", trust in system decreases (11), (14), (22). And trust decline is already not just reputational problem. It directly affects stream quality. People start sorting less carefully, mix organics with packaging, throw inappropriate items in containers or generally stop believing their efforts make sense. As result contamination of orange stream grows, recycling becomes more expensive and less efficient, and system enters even deeper into cycle of self-confirming inefficiency (11), (14), (22).

Precisely therefore the third section in this analytical paper is important not only as description of individual failures. Its meaning is different: it shows that TAMIR weakness lies not in one specific failure and not in private management error. Before us is system that knows how to produce convincing reporting, but significantly worse knows how to cover entire market, account for real material flows and bear responsibility for its peripheries - for those producers who did not enter system, for those fractions poorly recycled, and for those

areas where infrastructure either works poorly or actually does not work at all. As result official success proves built on exclusions, methodological narrowings and constant pushing most inconvenient elements beyond main showcase. Precisely this makes system not just imperfect, but structurally inefficient.

4. Institutional deadlock: TAMIR, landfills, and Cleanliness Fund

TAMIR system efficiency cannot be assessed in isolation, as if it existed by itself, separate from the entire rest of Israel's waste management economy. In practice, TAMIR is embedded in a much broader institutional construction where its operation is directly influenced by landfill operators, landfilling tariffs, and how the state disposes of Cleanliness Fund (קרן הניקיון) resources (5), (7), (27), (28). Precisely at this level it becomes especially clear that the problem lies not only in the corporation itself and not only in the quality of its operational model. Much more important is that around it has formed a system of distorted incentives in which waste landfilling too often remains a simpler, more predictable, and sometimes more economically convenient scenario than deep recycling (5), (7), (27), (28).

Formally the Israeli system should have been structured differently. Landfilling tax (היטל הטמנה) was introduced precisely as a tool to discourage landfills. Its logic is clear: the more expensive sending waste to landfill becomes, the stronger the incentive municipalities and operators should have to seek alternatives - develop sorting, recycling, separate collection, and reduce landfilling volumes (29), (30). Over the last ten years this tax has indeed noticeably grown and by 2024 reached approximately 115–120 ₪per ton of municipal waste (29), (31). On paper this looks like a serious step in the right direction. But in the real system one tariff increase proved insufficient because the tax began working not in empty space, but within already established and far from always rational infrastructure.

For municipalities this created an almost classic trap. On one hand, landfilling became more expensive due to growing tax. On the other hand - separate collection and recycling through TAMIR, which theoretically should have become profitable alternative, in practice remain expensive, organizationally complex, and dependent on constant additional payments from city budget. In other words, the city finds itself between two unfavorable options. Landfilling is expensive but at least remains logistically understandable and relatively predictable. Recycling through TAMIR looks formally compensated but in reality requires complex infrastructure, additional routes, manual control, constant support, and hidden municipal co-financing (6), (12), (25), (31). As result landfilling tax does not eliminate the problem but only increases tension within already non-functioning financial model.

Added to this is another important factor - so-called gate fees, i.e. tariffs for waste acceptance at landfills, transfer stations, and sorting facilities. These tariffs are set by infrastructure operators, many of which actually occupy monopoly or oligopoly position at regional level. In conditions when Israel still lacks modern sorting and recycling capacities, entry to recycling facilities often proves comparable in cost to regular landfilling (27), (29), (32). This undermines the very economic logic of transition to recycling. If price of entry into "more ecological" scenario turns out not lower but sometimes almost the same as landfill price, then

the system begins losing its main incentive to change behavior. In such configuration TAMIR gains no real competitive advantage relative to familiar landfilling infrastructure.

Particularly acutely this problem manifests in connection with Cleanliness Fund. By law precisely there should go funds from landfilling tax - that is, essentially money collected from industry and municipalities in exchange for promise they will be returned back to system as investments into recycling infrastructure and industry modernization (29), (30), (33). In reform logic this looks quite reasonable: if state makes landfilling more expensive, then collected funds should help create alternative - new sorting lines, recycling capacities, logistical hubs, infrastructure for difficult fractions, and other elements without which impossible to reduce landfill dependence.

However precisely here arises one of the most serious institutional failures. State Comptroller reports repeatedly pointed out that huge sums, including billions of shekels, accumulate for years on Fund accounts but do not turn into real infrastructure facilities (27), (33). In other words, money is extracted from the system but returns back in form of working solutions only partially and with great delay. For TAMIR this means almost paradoxical situation: corporation must demonstrate EPR success but does not have sufficient access to those state resources that could make this success materially possible. It collects packaging but cannot properly rely on modern national deep recycling infrastructure, especially when talking about plastics, glass, and other technologically complex fractions (1), (6), (32).

Essentially the system proves "toothless" not only due to own limitations but also because state investment circuit is not directly connected to it. Cleanliness Fund exists, money accumulates in it, but TAMIR as recognized operator does not receive from this automatic institutional reinforcement. Instead of linked system where landfilling tax reduces flow to landfills while Cleanliness Fund simultaneously finances recycling alternative, arises a gap: penalty signal exists, but full infrastructural support does not. Precisely therefore landfilling tax in Israeli conditions more often works not as circular economy driver but as additional financial pressure on municipalities when recycling remains structurally weak.

This gap is exacerbated also by how Cleanliness Fund funds are used in practice. In expert discussions and State Comptroller materials repeatedly raised question that part of these resources actually serves not accelerating recycling but covering broader state budget needs or directed to projects only indirectly related to packaging waste (27), (33), (34). For municipalities this looks particularly painful. They pay growing landfilling tax, bear increasingly high sanitation system costs, but do not see these money returning to them sufficiently in form of new capacities, cheaper logistics, or real reduction of city budget burden. In such situation not surprising that trust in very idea of tax as ecological tool begins eroding.

No less important is contractor level where institutional distortions become even more down-to-earth and therefore even more destructive. Most municipalities do not haul waste by own forces but work through private contractors - both for mixed waste and separate collection streams (6), (11), (32). And precisely here arises conflict of interests which TAMIR in current model can hardly control. For contractor with high operating costs and weak oversight too often proves more profitable to simplify route and take waste where acceptance is clear and

stable - i.e. toward mixed stream and landfill, especially if TAMIR sorting station is far, overloaded, or gives no obvious economic gain (11), (12), (32).

From this follows also chronic stream mixing problem. Even if formally municipality organizes separate collection, this still does not mean all collected material will really reach recycling in form assumed by official model. With lack of control, high transport costs, and insufficient contract transparency, contractor finds easier to combine orange container content with mixed waste or send significant stream part toward facilities where waste further fate proves less transparent (11), (12), (32). For population this looks like direct betrayal of sorting logic; for system - as yet another gap between official scheme and actual practice.

Even in cases when waste still reaches sorting station, the problem does not disappear. Significant stream part after sorting turns into so-called "residues" recognized unfit for recycling. According to various estimates their share can reach 40–60% depending on collected material quality and line technical capabilities (6), (21), (32). These "tails" then still go to landfilling. But costs of their haulage, placement, and landfilling tax payment far from always obvious in contract structure. Often they either directly fall on municipality or dissolve in opaque financial schemes between contractor, station operator, and city (6), (21), (32). As result even separate collection can end with same landfill - only through longer and more expensive path.

Added to this is monopoly problem. TAMIR status as sole recognized operator (גוף מוכר) formally should have ensured system manageability but simultaneously conserved its weaknesses (3), (4), (27). Unlike some European countries where several competing EPR operators work in EPR sphere, Israeli model creates no serious pressure toward cost reduction, new solutions search, or more aggressive work with evading producers (5), (27), (28). Monopoly gives stability but deprives system of internal development incentive. When operator is one and state extends recognition of its status, appears too much space for inertia and too few reasons to radically revise inconvenient model elements.

But perhaps even more important is that TAMIR does not exist separately from other waste market players but forced to act in field largely controlled by same transfer station and landfill operators for whom landfilling remains basic business economics (27), (32), (34). These players can formally participate also in recycling infrastructure but not interested in sharp waste flow reduction to landfills if precisely this flow ensures them main income. In other words, TAMIR proves embedded in environment where its success depends on actors whose economic interests far from always coincide with task of radical landfilling reduction. This is real institutional deadlock: system declares transition to circularity but relies on same structural beneficiaries of linear model for whom landfill still remains economy center.

Precisely therefore conflict between TAMIR, landfills, and Cleanliness Fund cannot be reduced to individual administrative failures or question of one operator insufficient efficiency. Before us deeper problem. State simultaneously tries making landfilling more expensive but does not create sufficient recycling alternative; accumulates funds in Cleanliness Fund but does not turn them into infrastructural leap; imposes on TAMIR responsibility for EPR results but leaves it within market dominated by players interested in

maintaining large landfilling volumes. In such system even well-conceived tools start working against each other.

In the end forms paradoxical but quite stable construction. Landfilling tax exists but does not break landfill dependence model. Cleanliness Fund exists but does not ensure timely and large-scale industry modernization. TAMIR exists as formal EPR flagship but receives neither sufficient infrastructural support nor market space where recycling would really prove more profitable than landfilling. Therefore institutional deadlock here lies not in absence of ecological tools as such, but in that they are not assembled into single working system. And while this fragmentation persists, recycling in Israel will remain not natural result of rational policy but weaker and more vulnerable contour compared to familiar landfill economics.

5. International context: Lagging behind Europe (PPWR)

TAMIR system and Israeli packaging regulation in official rhetoric often presented as movement toward "European EPR model" and circular economy (2), (5), (7). However if comparing currently operating in Israel construction with new European Packaging and Packaging Waste Regulation (PPWR), and also with how today in EU understood Extended Producer Responsibility, becomes clear that similarity here largely declarative. Essentially Israel still remains in logic of old packaging policy where main emphasis made on mass of collected and recycled material, while Europe ever more noticeably shifts to another side - toward waste prevention, reuse, eco-design, and eco-modulation of fees (5), (7), (20), (24), (35).

Precisely in this consists main international context for TAMIR assessment. Talking already not simply about whether Israeli system works well or poorly by itself. Question is to what extent it corresponds at all to direction in which developing modern regulatory logic of packaging management. And here gap proves quite substantial.

New PPWR, politically agreed in 2023–2024 years, changes very philosophy of European policy in this sphere (24), (35), (36). If earlier main theme was question how much packaging managed to collect and how much - recycle, then now center of gravity shifts higher up the chain. Europe demands not only better recycle waste but also in principle reduce packaging volume per capita, eliminate excessive elements, voids, and unnecessary multi-layer solutions, replace single-use formats with reusable where really possible, and make packaging design market entry condition (24), (35). In this same context reuse becomes ever more important: for range of sectors including HoReCa, delivery, and beverages introduced mandatory shares of reusable packaging and requirements to reuse systems (24), (35), (36). Finally EPR itself in European understanding ever less reduced to simple money collection for already produced packaging and ever more becomes tool which should preemptively influence producer behavior - through fee structure, design standards, and economic pressure toward more ecological solutions (20), (24), (35), (36).

Against this background Israeli model looks much more conservative. Packaging Law of 2011 and subsequent regulatory frameworks almost contain no eco-design requirements. System has no serious restrictions on multi-layer, hard-to-separate packaging, composites, or other solutions which a priori create sorting and recycling problems. No direct requirements

also to minimizing excessive packaging - extra boxes, nested plastic elements, decorative but functionally unnecessary layers (2), (6), (28), (37). In other words producer in Israel still can place on market packaging extremely inconvenient for subsequent recycling while his economic obligations in TAMIR system will not differ too strongly from obligations of producer of more rational and ecological packaging (21), (23), (28).

No less indicative also absence of mandatory reuse targets. In Israeli law no equivalent of those reuse targets which PPWR makes part of overall packaging policy. No quotas on reusable beverage containers beyond quite narrow deposit sector, no requirements to reusable packaging in delivery, e-commerce, or HoReCa spheres, no serious incentives created for return of older but from circularity perspective effective models including returnable glass containers (5), (7), (24), (28), (37). As result entire system remains focused mainly on how to collect and recycle already formed waste rather than how to make this waste arise less.

Here particularly clearly manifests limitation of TAMIR logic itself. Entire corporation operational model built around downstream stage: collect packaging through orange containers, sort it, partially recycle, and meet mass targets of declared packaging (1), (3), (6), (21). This undoubtedly important function. But it almost does not touch upstream level - i.e. packaging design, marketing decisions, material choice, producer business models, and very commodity market structure (5), (7), (28), (37). Essentially TAMIR services already existing packaging economy rather than transforms it. And in this sense Israel uses EPR more as consequence management mechanism than as industrial and ecological policy tool.

Special role here plays eco-modulation of tariffs question. In modern European EPR logic producer fees should be arranged to encourage easily recyclable or reusable packaging and conversely punish packaging creating increased ecological costs (20), (24), (35), (36). In theory Israeli system may seem moving in this direction: TAMIR indeed has different " דמי טיפול" rates for plastic, cardboard, metal, glass, and other materials (21), (23). But upon closer examination becomes clear that this differentiation remains too coarse and aggregated.

It differentiates primarily large material categories but almost does not react to differences within categories themselves. Meanwhile precisely there hide most important ecological differences: monomaterial or complex composite, exists in Israel real recycling infrastructure precisely for this fraction, exists market outlet for it, possible to replace with reusable alternative (21), (23), (28). Besides in system almost no strong mechanism directly rewarding mass and volume reduction of packaging as such. Although fee formally depends on mass it not converted into full regulatory preventive design tool, and therefore producer can still increase packaging without really tangible tariff pressure (2), (6), (21), (23). No less important also that such modulation methodology remains insufficiently transparent. In European countries ever more often published detailed principles of eco-modulated fee calculation; in Israel this connection between packaging ecological quality and producer financial burden remains noticeably less clear and less public (20), (24), (35), (36). As result signals producers receive from TAMIR tariffs prove weak and lagging. For most companies packaging optimization remains rather branding or marketing question than consequence of serious regulatory pressure (21), (23), (28).

Indicative also comparison of how in Europe and Israel relate deposit system and EPR. In range of EU countries deposit systems (DRS) do not exist as separate almost autonomous world but embedded in overall Extended Producer Responsibility architecture. There financial flows between deposit and EPR more coordinated, and profit from high-income fractions - primarily PET and aluminum - can be used to support loss-making streams like non-deposit glass or complex plastics (20), (24), (35), (36). State in such configuration acts not simply as regulator of several separate regimes but system architect balancing interests of operators, municipalities, and recyclers.

In Israel picture different. As already shown in first section beverage deposit system and TAMIR function as two weakly connected ecosystems. Here no legal mechanism redistributing DRS rent in favor of weaker EPR segments, and no holistic framework linking deposit targets with TAMIR targets and broader state waste strategy (15), (18), (19), (20), (24). From modern European logic perspective this means quite simple thing: Israel formally introduced both EPR and deposit but never turned them into single integrated packaging policy (20), (24), (35). And this already not simply technical difference but serious institutional lag.

Such lag has not only ecological but also politico-economic consequences. As PPWR deeper embeds in European regulatory practice retailers, brands, and importers in EU will start demanding from supplier packaging ever tougher requirements including design requirements, recyclability, recycled content, and packaging excessiveness degree (24), (35), (36), (38). For Israeli producers accustomed to softer domestic environment this creates risk of trade and reputational losses. In other words lag from European logic over time can turn not only into ecological problem but also into obstacle for foreign economic competitiveness.

No less important also missed industrial policy opportunities. In EU EPR and PPWR ever more often become tools of transition to "green" economy: stimulate recycling development, reuse services, eco-design, local technological solutions, and new value chains (5), (7), (24), (28), (37). In Israel current model on contrary largely conserves existing market structure: import of cheap single-use plastic, weak recycling, limited development of local reuse and deep recycling sectors (5), (7), (24), (28), (37). System continues working in waste management logic rather than material economy restructuring logic.

Finally there is also longer-term dimension - growth of ecological debt. The longer Israel postpones transition from "tons recycled" logic to prevention, reuse, and eco-design logic, the more complex and expensive subsequent leap will prove (5), (7), (28), (34). For country with limited territory, high population density, and already accumulated landfill dependence damage this particularly important. In other words today's institutional lag is not simply question of mismatch with European fashion or rhetoric. This question of how long country can afford postponing transition to tougher and more modern packaging management model without accumulating even larger financial, ecological, and infrastructural costs.

Precisely therefore international context in this paper important not as external background but as kind of mirror in which particularly clearly visible current TAMIR model limitations. If judging only by internal reports Israel may look like country where EPR already works and gives high indicators. But if looking at system through PPWR and modern European practice

prism becomes clear talking about much narrower and obsolescing construction. It knows how to count collected and recycled tons but significantly worse knows how to prevent packaging formation, influence its design, promote reuse, and coordinate different extended responsibility regimes into single policy. And therefore problem here not only in that Israel so far not caught up with Europe. Problem in that TAMIR architecture itself still embedded in yesterday's regulatory logic while tomorrow's requirements already formulated quite clearly.

Conclusions and recommendations: Path to real responsibility

Conducted analysis shows that current TAMIR model in Israel works not as full Extended Producer Responsibility system but rather as partial cost compensation mechanism where significant part of financial and operational burden ultimately remains on municipalities and taxpayers. Therefore for exit from this institutional deadlock insufficient to improve reporting or slightly adjust tariffs. Needed deeper reconfiguration of entire packaging waste management architecture - one reducing "invisible subsidy", eliminating incentive distortions, and shifting system from quantitative indicators demonstration to real structural transformation (5), (6), (27), (28), (35).

Financial model reform

First step here should become reform of very financial model itself. While TAMIR payments built on averaged standards rather than real city logistics cost, municipal deficit will reproduce again and again (12), (25), (31). Therefore logical to transition from single national compensation rate to more flexible system accounting for actual differences between cities: complex terrain of Haifa, Tel Aviv building density, periphery remoteness, varying last mile cost and container site servicing (6), (12), (25). No less important also include in compensation formula not only haulage as such but also city administrative expenses - contract management, contractor monitoring, inspections, complaint processing, cleaning around accumulation points. In other words pay not only for ton but also for system making this ton manageable (25), (26). Finally tariffs themselves should index automatically - taking into account landfilling tax growth, fuel prices, and sanitation wages so municipalities not become hostages of inflation and annual manual revisions (29), (31).

Deposit-EPR integration

But one financial correction insufficient if gap between deposit system and EPR preserved. Today these two regimes exist side by side but do not form overall balance: deposit contour concentrates profitable fractions while TAMIR and cities receive mainly expensive and problematic remainder (15), (19), (20). Precisely therefore necessary end this kind of economic cannibalism. One possible path - creation of single clearing center linking TAMIR and deposit operators financial flows. In such model income from highly liquid raw materials - primarily PET and aluminum as well as part of unclaimed deposits - could direct toward supporting recycling of fractions remaining chronically loss-making themselves: non-deposit glass, complex plastics, mixed materials (20), (24), (35). In parallel needed also unified national KPIs uniting both systems around common goal - not fighting for individual bottles

but reducing waste as such and increasing overall packaging policy sustainability (24), (35), (36).

True eco-modulation

Deserves separate attention question of genuine eco-modulation. Israel long approached point when TAMIR tariffs should cease being simply fiscal tool and turn into industrial and ecological design instrument as happens in EU within PPWR logic (24), (35), (37). This means packaging actually impossible or extremely difficult to recycle should taxed with multiple higher fees. Talking about multi-layer materials, black plastic, complex composites, and other solutions a priori creating excess costs for entire system (20), (35), (37). But one punitive measure here insufficient. Needed also reverse incentives - direct financial bonuses for producers transitioning to monomaterials, reducing packaging mass, implementing eco-design solutions or developing reuse models and reusable containers (35), (36), (37). Meanwhile calculation methodology itself should become much more transparent. While connection between fee size and real recycling cost of specific packaging type under Israeli conditions remains too opaque, market signals will be weak and lagging (21), (23), (28).

Data transparency

No less critical also data transparency. Without it impossible neither honestly assess problem scale nor force system work by rules. Today fight against "free riders" and statistical distortions remains one of main weaknesses of entire mechanism (6), (8), (11). Therefore necessary single digital platform linking Customs Service, Tax Authority, and TAMIR data and allowing automatic detection of companies not fulfilling Packaging Law (6), (9), (10). Simultaneously should change also very principle of public reporting: Ministry of Environmental Protection should publish recycling indicators calculated not only from TAMIR declared base but from total packaging volume on market including free riders segment assessment (8), (11), (14). And finally system needs regular independent audit - not formal but based on morphological waste composition studies at landfills and other facilities conducted by external scientific organizations. Only such verification will allow check how much reporting corresponds to material reality (27), (28), (34).

Unlocking Cleanliness Fund

Separate reform direction should become unlocking Cleanliness Fund. While it functions rather as black box than modernization driver system will suffocate from infrastructure shortage (27), (30), (33). Accumulated there resources should not lie dead weight but turn into real facilities: modern automated sorting centers, deep recycling lines, capacities for separating complex plastics and other difficult fractions (32), (33). But this also insufficient if secondary raw materials have no stable domestic demand. Therefore part of funds should go also toward supporting local enterprises using recycled glass, plastic, and other materials so Israeli system not depend entirely on world oil price fluctuations and external markets (27), (32), (34). In other words Cleanliness Fund should become not passive money accumulator but real catalyst of infrastructure and market.

Waste prevention shift

Finally Israel necessary go beyond very "waste collection" logic and start seriously managing its formation. While policy focused almost exclusively on downstream stage it doomed constantly fight consequences rather than problem source (24), (35), (38). From this follows also another set of recommendations: introduce mandatory reusable packaging share targets in food delivery, e-commerce, and beverages sectors as well as fix standards limiting excessive packaging use - primarily where container volume clearly disproportionate to goods volume (24), (35), (36), (37). Such measures important not only themselves. They would mean transition from old paradigm where state merely tries better utilize already produced waste to new where it starts influencing design, market decisions, and very material consumption structure.

In the end main conclusion of this paper consists in that TAMIR system in its current form apparently already approached limit of its effectiveness within old paradigm. Further increasing "paper successes" possible but ever harder turn them into real ecological results. Without radical reform - financial, institutional, and regulatory - aimed at eliminating municipal subsidy, integrating with more modern PPWR logic, and shifting emphasis from compensation to prevention Israel risks remain country of high report percentages and growing landfill waste mountains. Realization of "polluter pays" principle requires not cosmetic improvements but political will: transfer financial responsibility from citizens and cities back to producers and turn EPR from limited fiscal mechanism into tool of ecological economy modernization.

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